



**Parul University**  
**Faculty of Engineering and Technology**  
**Parul Institute of Engineering and Technology**  
**Computer Science & Engineering Department**

|              |                                                                                                                 |          |                 |            |
|--------------|-----------------------------------------------------------------------------------------------------------------|----------|-----------------|------------|
| Subject Name | Data Analytics and Data visualization                                                                           | A.Y      | 2026-27         |            |
| Subject Code | 303105314                                                                                                       | Semester | 5 <sup>th</sup> |            |
| Assignment-1 |                                                                                                                 |          |                 |            |
| Sr No        | Question                                                                                                        | COs      | B.T             | Competence |
| 1.           | Explain the fundamentals of data analytics lifecycle and the role of data visualization in decision making.     | CO1      | L2 – Understand | 1          |
| 2.           | Analyze different types of data and measurement scales used in data analytics applications.                     | CO1      | L4 – Analyze    | 2          |
| 3.           | Compare various data analytics techniques and visualization methods using Python tools.                         | CO2      | L4 – Analyze    | 3          |
| 4.           | Examine how data generated from different sources adds business value with suitable real-world examples.        | CO1      | L4 – Analyze    | 4          |
| 5.           | Evaluate the importance of data analytics in improving organizational performance and strategic planning.       | CO1      | L5 – Evaluate   | 5          |
| 6.           | Differentiate between data analysis and data analytics with respect to objectives, processes, and applications. | CO2      | L4 – Analyze    | 2          |
| 7.           | Illustrate the elements and workflow of a data analytics system using suitable examples.                        | CO1      | L3 – Apply      | 3          |
| 8.           | Compare the roles, responsibilities, and skill sets of a Data Analyst and Data Scientist.                       | CO2      | L4 – Analyze    | 6          |
| 9.           | Apply probability-based testing methods and statistical techniques for data analysis problems.                  | CO3      | L3 – Apply      | 7          |
| 10.          | Apply regression, classification, and clustering techniques to solve real-world analytical problems.            | CO4      | L3 – Apply      | 3          |